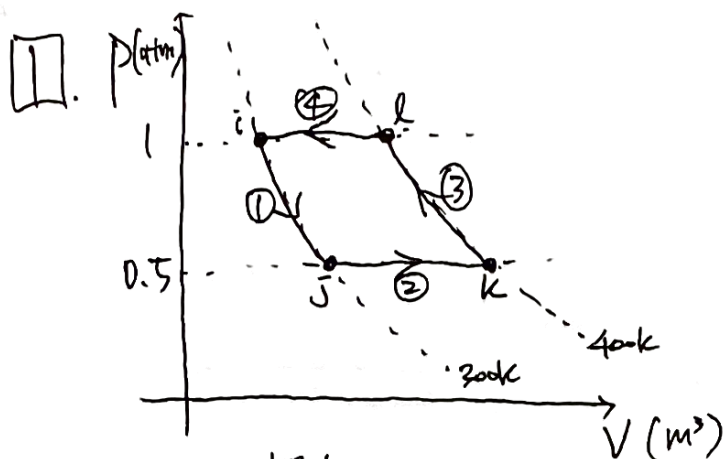




(1) ①: $W = \int_{V_i}^{V_f} P dV = nRT \int_{V_i}^{V_f} \frac{dV}{V} = nRT \ln \frac{V_f}{V_i}$

(20)

$$\left(\begin{array}{l} PV = nRT_{const} \\ P_i V_i = P_j V_j \\ \frac{V_f}{V_i} = \frac{P_i}{P_j} = \frac{1}{0.5} = 2 \end{array} \right)$$

$$W = 1 \text{ mol} \times 8 \text{ J/mol}\cdot\text{K} \times 300 \text{ K} \times \ln 2 = 1680 \text{ J}$$

④

$$\begin{aligned} \text{②: } W &= \int_j^k P dV = P (V_k - V_j) = P \left(\frac{nRT_k}{P_k} - \frac{nRT_j}{P_j} \right) = nR(T_k - T_j) \\ &= 1 \text{ mol} \times 8 \text{ J/mol}\cdot\text{K} \times (400 - 300) \text{ K} \\ &= 800 \text{ J} \end{aligned}$$

④

$$\begin{aligned} \text{③: } W &= \int_{V_k}^{V_l} P dV = nRT \ln \frac{V_l}{V_k} \\ &= 1 \text{ mol} \times 8 \text{ J/mol}\cdot\text{K} \times 400 \text{ K} \times \ln \frac{1}{2} \\ &= -2240 \text{ J} \end{aligned}$$

④

$$\begin{aligned} \text{④ } W &= \int_l^i P dV = nR(T_i - T_l) = 1 \text{ mol} \times 8 \text{ J/mol}\cdot\text{K} \times (300 - 400) \text{ K} \\ &= -800 \text{ J} \end{aligned}$$

④

$$\text{total } W = 1680 + 800 - 2240 - 800 = -560 \text{ J}$$

②

$$V_a = \frac{nRT_i}{P_i} = \frac{1 \text{ mol} \times 8 \text{ J/mol}\cdot\text{K} \times 300 \text{ K}}{10^5 \text{ N/m}^2} = \frac{2400 \text{ J}}{10^5 \text{ N/m}^2} = 0.024 \text{ m}^3$$

②